

HTTP Requests Analyzer

1. Module Overview

This module will provide a service that will analyze HTTP requests and provide statistics about them.

2. High Level Architecture

The application will be designed to be able to provide statistics given a log file, it will now support location, OS and browser statistics for an Apache HTTP requests but will be designed to be extended and support other statistics and different request logs.

3. Main Objects

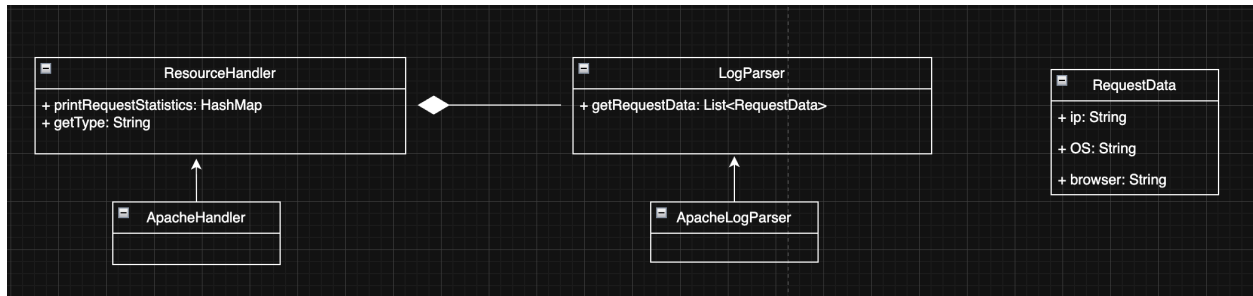
First, I will define a Handler interface that will implement different handlers, each will handle a different resource (For example, Apache) and will extract its output and print it.

Each handler will hold a LogParser by composition which will be responsible to the actual extraction of the data from file, the log parsers will get the raw data and return a list of "RequestData" which will be the data extracted for the application to work with.

All of the handlers and parsers will use the same services to parse and extract the correct data, the services will be:

- GeolpService – responsible to get the country using the IP.
- UserAgentService – responsible to get the browser from the user agent data.
- OutputService – responsible to print out the output.
- AnalyzerService – responsible to extract the statistics from the data.

This architecture is easily expandable to add new resources or to add new data to extract.



4. Additions I would have added given more time

- An LRU cache to reduce DB queries while extracting the IP to Country data.
- Implement logic to support automatic recognition of the resource an initialization of the correct handler.